PROBLEM 6 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Linear and Planar Densities in the [111] Direction and (111) Plane of Fe (BCC) and Ni (FCC)

Original statement · Problems T2_CrystStruct.pdf

6. Calculate the linear density in the $[111]$ *direction and the planar density in the* (111) *plane for:* ***a)*** *Iron BCC (Solution:* $\rho_{[111]} = 2/(\sqrt{3}a)$*,* $\rho_{(111)} = 1/(a^2\sqrt{3})$*)* ***b)*** *Nickel FCC (Solution:* $\rho_{[111]} = 1/(a\sqrt{3})$*,* $\rho_{(111)} = 4/(a^2\sqrt{3})$*)*

Solution: $\rho_{[111]} = 2/(\sqrt{3}a)$, $\rho_{(111)} = 1/(a^2\sqrt{3})$

PHASE 1: HYPOTHESIS & PARAMETERS

Structural Models:

1. **Ferritic Iron (α-Fe):** Body-Centred Cubic (BCC) lattice, parameter a_{Fe}, $n = 2$ atoms/cell.
2. **Nickel (Ni):** Face-Centred Cubic (FCC) lattice, parameter a_{Ni}, $n = 4$ atoms/cell.

Comparison Parameters:

- Direction evaluated: $[111]$ (main cube diagonal, length $L_{[111]} = a\sqrt{3}$).
- Plane evaluated: (111) (plane that intersects the axes at $x = a, y = a, z = a$, bounding an equilateral triangle of edges $a\sqrt{2}$).

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the lattice packing theory [Session 4 Slides 7-9]:

- **Linear Density:**

$$\rho_{[111]} = \frac{N_{\text{atoms centred on the diagonal}}}{a\sqrt{3}}$$

- **Planar Density:**

$$\rho_{(111)} = \frac{N_{\text{atoms contained in the triangle}}}{A_{\text{triangle}}}$$

where the triangle on the cell has edge $L = a\sqrt{2}$ and area:

$$A_{\text{triangle}} = \frac{\sqrt{3}}{4}L^2 = \frac{\sqrt{3}}{4}(2a^2) = \frac{\sqrt{3}}{2}a^2$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

Part a: BCC Iron (α-Fe)

1. **Linear Density on** $[111]$**:**
2. In BCC, atoms are in contact along the main diagonal $[111]$.
3. The diagonal passes through the two opposite vertices and goes through the central body atom.
4. Atoms cut: $2 \times \frac{1}{2}(\text{vertices}) + 1(\text{centre}) = 2$ atoms.
5. Segment length: $L = a\sqrt{3}$.

$$\rho_{[111]} = \frac{2}{a\sqrt{3}} = \boxed{\mathbf{\frac{2}{\sqrt{3}a}}}$$

6. **Planar Density on** (111)**:**
7. The (111) plane contains only the 3 vertices $(a,0,0), (0,a,0), (0,0,a)$. Each vertex contributes an angle of 60° inside the equilateral triangle ($\frac{60^\circ}{360^\circ} = \frac{1}{6}$):

$$N_{\text{atoms}} = 3 \times \frac{1}{6} = \frac{1}{2} \text{ atom}$$

- The central atom $(a/2, a/2, a/2)$ is **not contained in the** (111) **plane**, but lies at a perpendicular distance $d = \frac{a}{2\sqrt{3}}$ behind it.
- Area of the triangle: $A = \frac{\sqrt{3}}{2}a^2$.

$$\rho_{(111)} = \frac{1/2}{\frac{\sqrt{3}}{2}a^2} = \boxed{\mathbf{\frac{1}{a^2\sqrt{3}}}}$$

Part b: FCC Nickel (Ni)

1. **Linear Density on $[111]$:**
2. In FCC, atoms do not touch along $[111]$; the cube diagonal is empty at its centre.
3. The diagonal cuts only the two atoms at the opposite vertices through their centres:

$$N_{\text{atoms}} = 2 \times \frac{1}{2} = 1 \text{ atom}$$

- Length: $L = a\sqrt{3}$.

$$\rho_{[111]} = \frac{\mathbf{1}}{\mathbf{a\sqrt{3}}}$$

4. **Planar Density on (111):**
5. In FCC, the (111) plane is the plane of maximum packing (close-packed plane).
6. It contains the 3 vertices of the triangle (each contributes $\frac{1}{6}$) plus the 3 face centres located at the midpoints of the triangle sides (each contributes $\frac{1}{2}$ since it belongs to two adjacent cells):

$$N_{\text{atoms}} = 3 \times \left(\frac{1}{6}\right) + 3 \times \left(\frac{1}{2}\right) = \frac{1}{2} + \frac{3}{2} = 2 \text{ atoms}$$

- Area of the triangle: $A = \frac{\sqrt{3}}{2}a^2$.

$$\rho_{(111)} = \frac{2}{\frac{\sqrt{3}}{2}a^2} = \frac{\mathbf{4}}{\mathbf{a^2\sqrt{3}}}$$

`PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION`

1. Summary and mechanical consequences

Quantity	BCC iron (α-Fe)	FCC nickel (Ni)	Physical conclusion
Linear density $\rho_{[111]}$	$\frac{2}{a\sqrt{3}} \approx \frac{1.155}{a}$ (atoms in contact)	$\frac{1}{a\sqrt{3}} \approx \frac{0.577}{a}$	$\langle 111 \rangle$ is the slip direction of BCC
Planar density $\rho_{(111)}$	$\frac{1}{a^2\sqrt{3}} \approx \frac{0.577}{a^2}$	$\frac{4}{a^2\sqrt{3}} \approx \frac{2.309}{a^2}$ (close-packed)	$\{111\}$ is the slip plane of FCC

- The coefficients are compared for the same symbol a; iron and nickel have different lattice parameters, which the statement does not give, so the absolute densities of the two metals cannot be ranked from these formulas alone.
- In **BCC iron** the atoms touch along the cube diagonal ($4R = a\sqrt{3}$ [Session 4 Slide 5]), so $[111]$ is the close-packed direction, but its (111) plane is sparse. The densest BCC plane is (110), with $\rho_{(110)} = \sqrt{2}/a^2 \approx 1.414/a^2$ (Problem 4), and BCC slips on $\{110\}\langle 111\rangle$ [Session 4 Slide 37].
- In **FCC nickel** the (111) plane is the close-packed plane, four times denser (for equal a) than the BCC (111) plane, while the FCC close-packed direction is $\langle 110\rangle$, not $\langle 111\rangle$: FCC slips on $\{111\}\langle 110\rangle$ [Session 4 Slide 37].

2. Checks

- **Contact in BCC:** the close-packed linear density must be one atom per diameter, $\frac{1}{2R}$. With $2R = \frac{a\sqrt{3}}{2}$: $\frac{1}{2R} = \frac{2}{a\sqrt{3}}$, equal to $\rho^{\mathrm{BCC}}_{[111]}$.
- **Lattice repeat in FCC:** the point (t, t, t) on the diagonal is an FCC site only for integer t (corners); face centres have one coordinate different from the other two. The repeat distance is therefore the whole diagonal $a\sqrt{3}$, so $\rho^{\mathrm{FCC}}_{[111]} = 1/(a\sqrt{3})$.
- **Volume-density cross-check (extension):** $\rho_{(111)} = \rho_v \cdot d$, with d the spacing of consecutive atomic (111) layers. FCC: $\rho_v = 4/a^3$, layers $x + y + z = 0, 1, 2, 3$, $d = a/\sqrt{3}$, so $\rho_v d = \frac{4}{a^3} \cdot \frac{a}{\sqrt{3}} = \frac{4}{a^2\sqrt{3}}$. BCC: $\rho_v = 2/a^3$, layers $x + y + z = 0, 1, \frac{3}{2}, 2, 3$, $d = \frac{a}{2\sqrt{3}}$, so $\rho_v d = \frac{2}{a^3} \cdot \frac{a}{2\sqrt{3}} = \frac{1}{a^2\sqrt{3}}$.
- **Dimensions:** $[\rho_{[111]}] = \mathrm{atoms/m}$ and $[\rho_{(111)}] = \mathrm{atoms/m^2}$ for a in m .

3. Comparison with the official answer

All four results agree with the official solution: BCC $\rho_{[111]} = 2/(\sqrt{3}\,a)$, $\rho_{(111)} = 1/(a^2\sqrt{3})$; FCC $\rho_{[111]} = 1/(a\sqrt{3})$, $\rho_{(111)} = 4/(a^2\sqrt{3})$.